SE 441 – Software Engineering
Professional Ethics

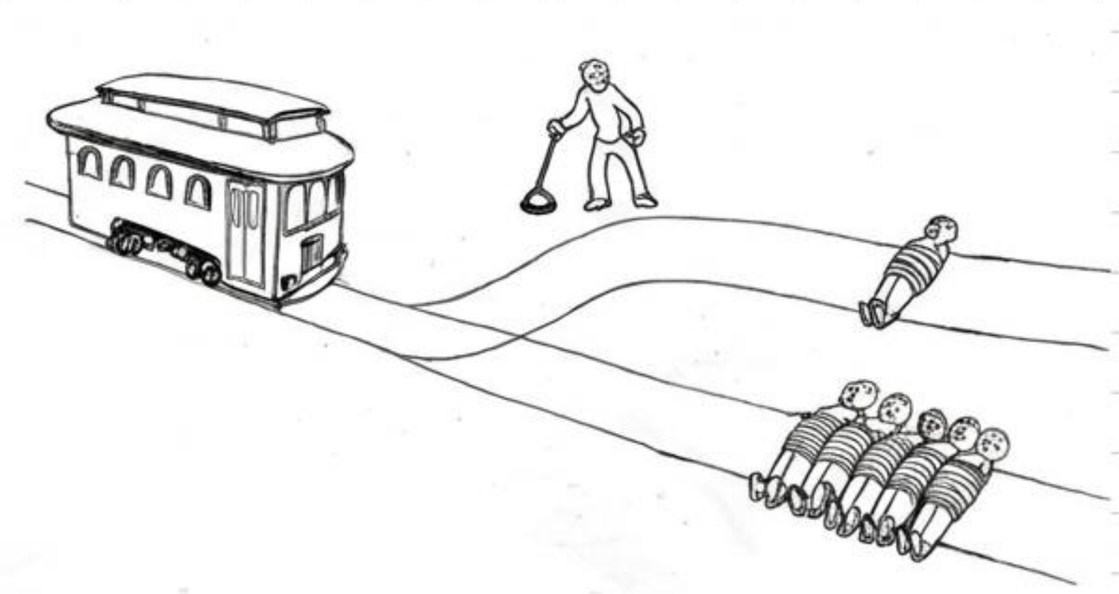
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The Math is Identical. The Choice is Not.

Scenario A



Would you pull the lever?

(1 dies, 5 live)

[DISCUSSION]

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The Server Room Trolley Problem

Release Dilemma

```
if (ship_early):
  Ship the medical software
  with a known 1% critical
  failure rate.
else:
  Delay the launch, bankrupt
  your startup, fire 50
  employees.
```

Privacy Dilemma

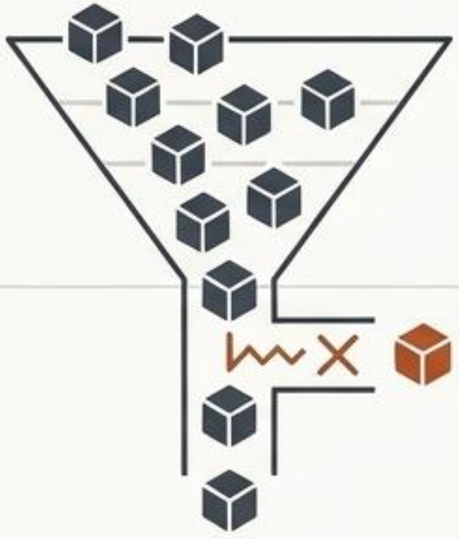
```
if (scrape_data):
  Secretly scrape private
  user health data to train a
  life-saving AI cure.
else:
  Protect user privacy, but
  the cure takes 10 years
  longer.
```

[DISCUSSION]

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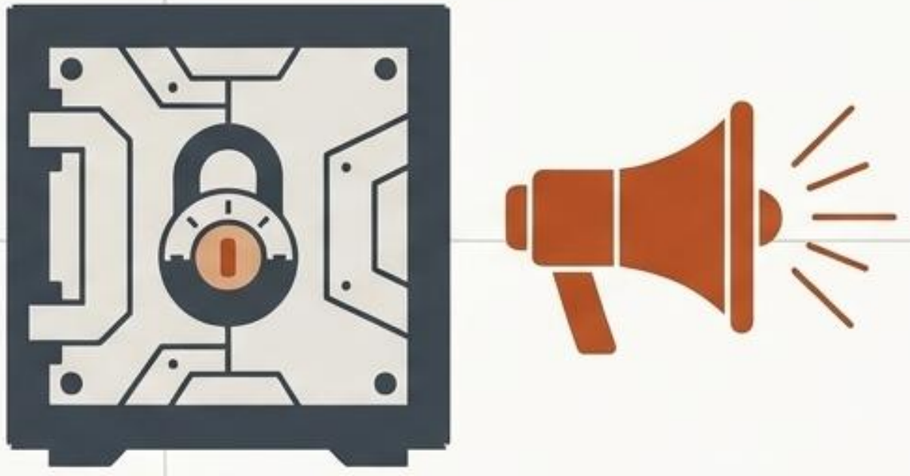
Accuracy vs Fairness | Loyalty vs Truth

AI Bias



Your new hiring algorithm is 15% more accurate for the company, but it systematically screens out a specific demographic. Do you deploy it?

Whistleblowing



You discover your employer is covering up a massive data breach. Do you break your strict NDA to alert the public?

[DISCUSSION]

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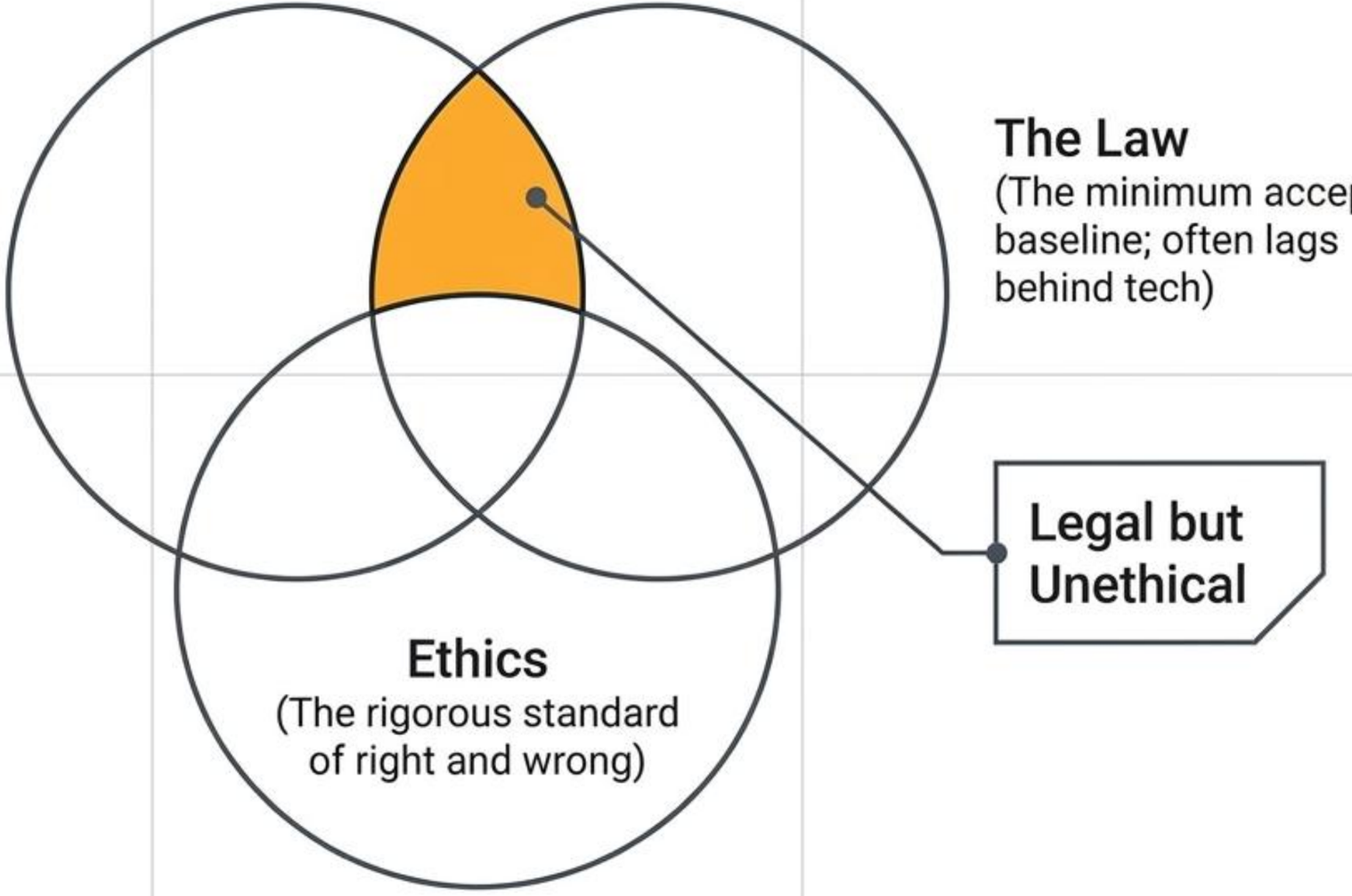
Intuition Doesn’t Scale.

You are all highly rational, intelligent engineers, yet you violently disagreed on the previous slides. When writing software for millions, we cannot rely on gut feelings. We need a rigorous framework.

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The Forces Governing Behavior

Personal Preference
(Arbitrary, individual)



The Law
(The minimum acceptable baseline; often lags behind tech)

Ethics
(The rigorous standard of right and wrong)

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Good Person vs. Good Engineer



Professionals possess asymmetric power. You understand the source code; the user does not. This knowledge creates a **higher standard of duty**.

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Why Software Needs Ethics

```
1 line of code × 1,000,000 users = Massive Impact
```



Scale:

A bridge collapse kills hundreds. A software failure crashes global markets or disrupts pacemakers.



Permanence:

Data collected today exists forever.



Opacity:

Black-box algorithms hide discrimination behind complex math.

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The Core Frameworks (Selecting a Paradigm)



Deontology (Duty)

Focus: The Rules & Actions.

Core Question: What is my absolute duty?



Utilitarianism (Outcome)

Focus: The Math & Consequences.

Core Question: What maximizes the good for the most people?



Virtue Ethics (Character)

Focus: The Habit & Intent.

Core Question: What would an excellent engineer do?

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Deontology — The Unbreakable Rule



Kant’s Categorical Imperative:

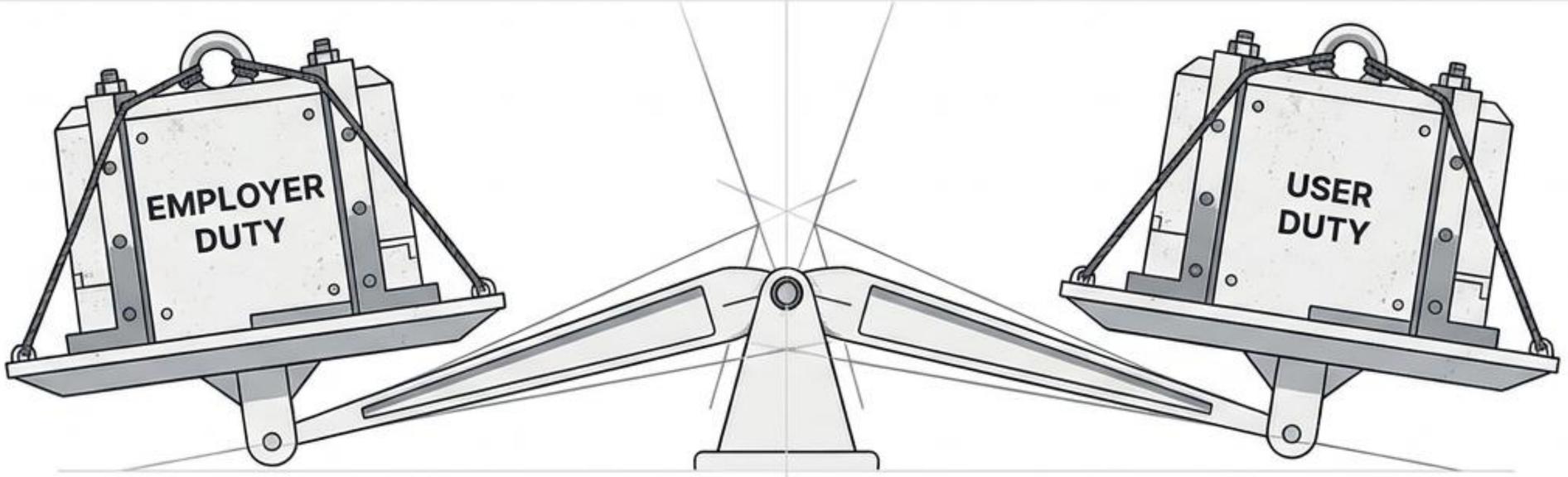
An action is only right if you would be willing to make it a universal law for everyone.

The SE Translation:

Never use unauthorized data.
Even if it builds a life-saving model.
The rule cannot be broken regardless of the consequence.

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The Collision of Duties



Duty to Employer	Duty to Users
Protect profits, honor NDAs, ship code fast	Protect privacy, ensure safety, provide transparency

Scenario: Your manager orders you to build a Dark Pattern into the UI to make users accidentally subscribe to a premium tier. Where does your duty lie? To your paycheck, or to your user?

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The Great Debate

The Scenario:
The government demands a **backdoor** in your secure messaging app to catch a known terrorist cell.

Action-Oriented (Deontology)

Argue why you MUST NOT build the backdoor.

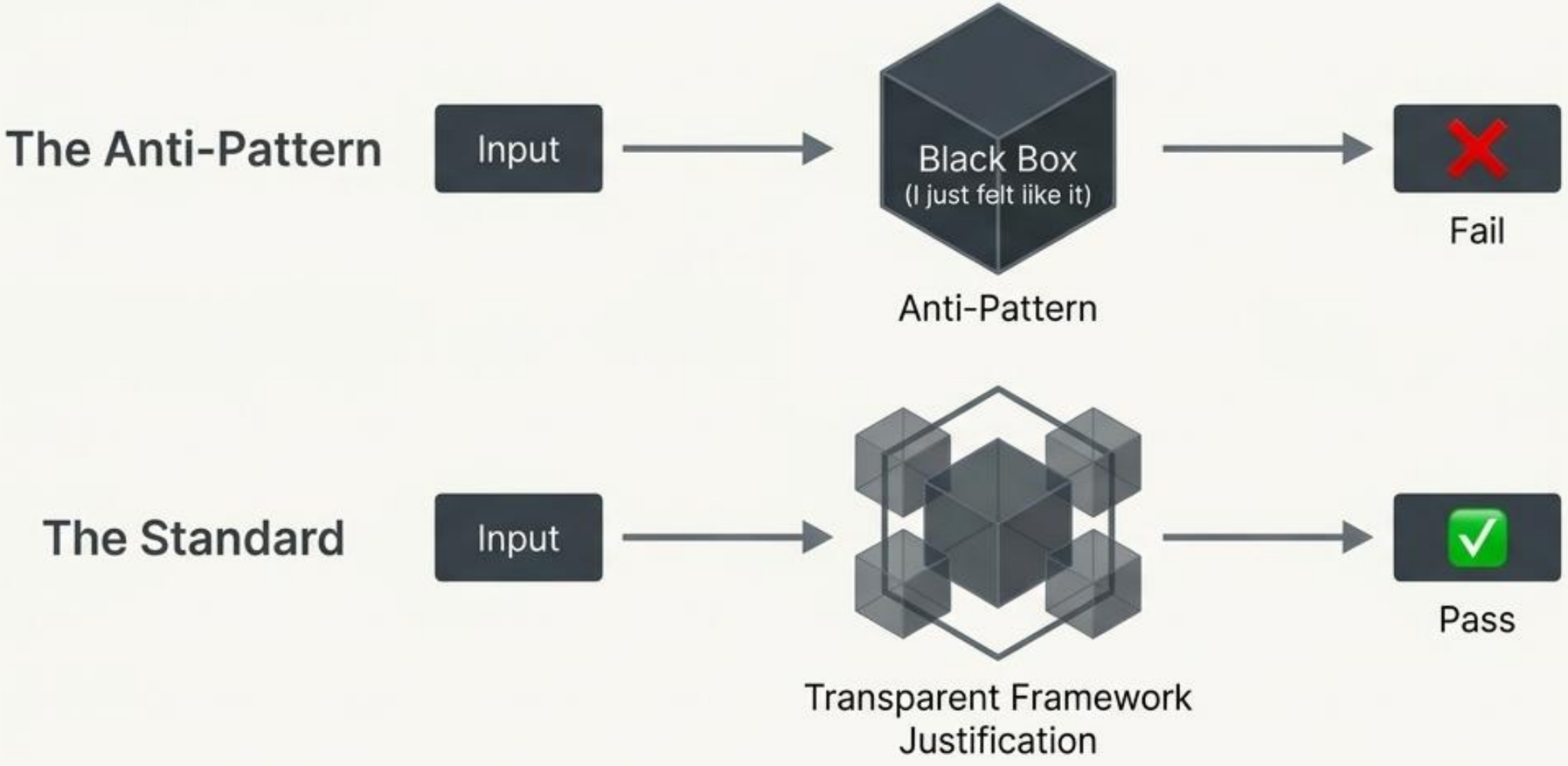
Consequence-Oriented (Utilitarianism)

Argue why you MUST build the backdoor.

[DISCUSSION]

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The Anchor of Justification

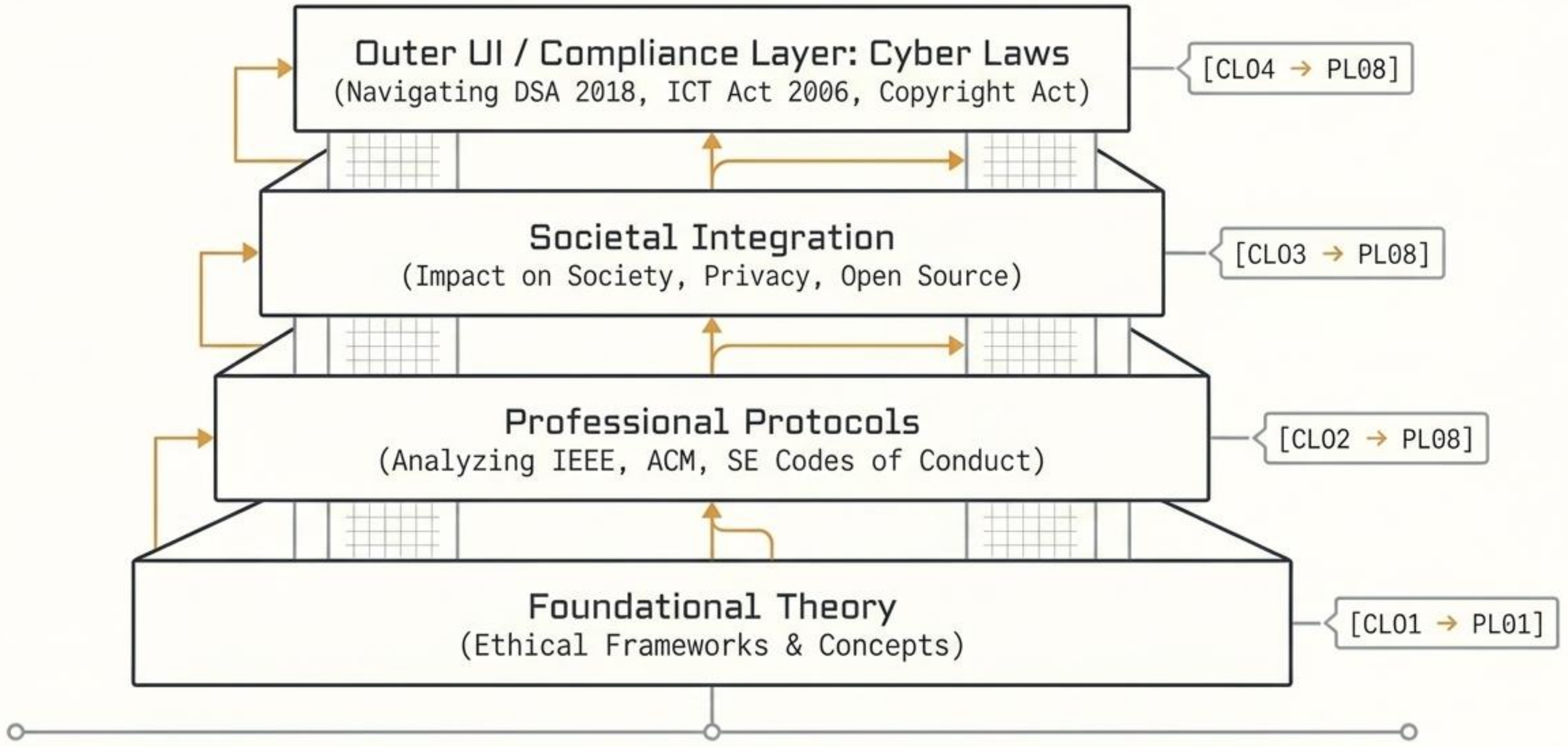


In software engineering, if your code breaks, you must be able to show your logic.

If your ethical decision causes harm, you must be able to justify it using established frameworks.
I felt like it is a failing grade.

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SE 441 System Architecture (Course Roadmap)



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Based on Slide C

The Fork in the Machine



What does an ethical software engineer do?

Think back to the trolley. Think back to the release dilemma.

Now that you see the underlying code of human decision-making… has your answer changed?

[DISCUSSION]

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